Baseline appendix

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A Baseline comparisons

The first two tables report Spearman correlations with LRE faithfulness across the LRE Dataset.

Table 1: **Readout baselines across seven models.** Spearman correlation ρ with LRE faithfulness.

Model	Differential Jacobian	Raw Jacobian	Logit lens	Differential logit lens	Attribute rate
OLMo 3 7B	0.683	**0.689**	0.234	0.326	−0.114
GPT-J 6B	**0.573**	0.567	0.531	0.394	0.002
Mistral 7B	**0.606**	0.586	0.402	0.249	−0.304
Gemma 2 9B	0.554	**0.575**	0.554	0.221	−0.292
Qwen3 8B Base	0.729	0.736	**0.743**	0.607	0.052
Qwen3.8 27B	0.600	**0.644**	0.402	0.454	−0.196
Pythia 6.9B[†]	0.479	**0.501**	−0.343	0.217	0.096

Bold marks the largest point estimate within each model, not a significant difference. [†]Pythia's attribute rate excludes 413 test rows with no predictions surviving the attribute filter; the other models have no undefined rates.

Readout definitions. For subject state s_e, reference states s_i, and reference Jacobians J_i, the differential and raw Jacobian readouts decode $\mathrm{mean}_i\, J_i(s_e - s_i)$ and $\mathrm{mean}_i\, J_i s_e$, respectively. Logit lens decodes s_e; differential logit lens decodes $s_e - \mathrm{mean}_i\, s_i$. Each relation's score is the negative median $\log_{10}(\mathrm{rank} + 1)$ of the model's answer token.

Attribute rate. Following Geva et al. (2023), we measure overlap between filtered top-50 logit-lens predictions and subject attributes retrieved from Wikipedia. The index uses the union of current test subjects; the source stopword, length, and retrieval filters are retained. Relation scores average the percentage overlap. Undefined rates are handled as noted in the table; matched-row sensitivity results remain in the accompanying data.

Raw Jacobian has a larger point estimate than differential Jacobian in five models. All seven paired 95% relation-bootstrap intervals for their difference include zero; these results do not establish a general advantage for either readout. LRE hyperparameter selection remains in sample.

A.1 OLMo corpus-frequency baselines

Table 2: **Frequency baselines for OLMo 3 7B.** Correlations with LRE faithfulness using document counts from OLMo's three published training mixes combined.

Frequency measure	Spearman ρ	p-value
Marginal frequencies		
Subject frequency, c_s	0.290	0.160
Object frequency, c_o	0.644	5.08×10^{-4}
Co-occurrence		
Joint frequency, c_{so}	0.578	0.002
Mean of $\log(1 + c_{so})$	0.568	0.003
$\log(1 + \text{mean}(c_{so}))$	0.578	0.002
Conditional frequencies		
Forward conditional, $P(o \mid s)$	0.386	0.056
Reverse conditional, $P(s \mid o)$	-0.328	0.109
Directional difference, $P(o \mid s) - P(s \mid o)$	0.362	0.075
Micro-averaged $P(o \mid s)$	0.615	0.001
Micro-averaged $P(s \mid o)$	0.030	0.887

Counts denote binary document presence. Means use the current test rows with both marginals observed; micro averages are ratios of summed counts. The p-values are two-sided and unadjusted.

Counting and aggregation. A document contributes at most one mention per subject/object string. Matching uses normalized literal names with Unicode word boundaries, without alias resolution or a proximity constraint. With subject, object, and joint counts c_s, c_o, and c_{so}, the conditionals are $P(o \mid s) = c_{so}/c_s$ and $P(s \mid o) = c_{so}/c_o$. Macro features average row-level quantities where both marginals are observed. Micro conditionals divide summed joint counts by summed marginals. The log of the mean joint count has the same Spearman correlation as the mean joint count because the transformation is monotonic.

Interpretation and reproducibility. These are exploratory associations. Relations can share subjects and objects, and the reported p-values are unadjusted. Conditional direction also reflects marginal frequency: when $c_{so} > 0$, $P(o \mid s)/P(s \mid o) = c_o/c_s$.

A.2 Neutral-prompt variations

Table 3 collects the prompt experiments for the differential Jacobian lens. For Qwen3 8B Base, we compare the original prompt with six alternatives and the subsequently measured newline control. The checkpoint, layer 19, reference and test rows, relation-prompt answer tokens, and LRE outcomes are fixed.

Table 3: **Neutral-prompt variations.** Spearman ρ between differential-Jacobian scores and retained LRE faithfulness.

Qwen3 8B Base: prompt wording and prefix

Neutral prompt	ρ
`{s} is known for`	0.268
`{s}`	−0.057
`{s} is`	0.166
`{s} can be described as`	0.372
`Information about {s}:`	0.733
`Facts about {s}:`	0.788
`Entity: {s} is known for`	0.728
`\n{s} is known for`	0.729

References

Mor Geva, Jasmijn Bastings, Katja Filippova, and Amir Globerson. Dissecting recall of factual associations in auto-regressive language models. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12216–12235, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.751. URL https://aclanthology.org/2023.emnlp-main.751/.